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LITERATURE REVIEW

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Computational modeling of whole-brain dynamics: a review of neurosurgical applications

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A major goal of modern neurosurgery is the personalization of treatment to optimize or predict individual outcomes. One strategy in this regard has been to create whole-brain models of individual patients. Whole-brain modeling is a subfield of computational neuroscience that focuses on simulations of large-scale neural activity patterns across distributed brain networks. Recent advances allow for the personalization of these models by incorporating distinct connectivity architecture obtained from noninvasive neuroimaging of individual patients. Local dynamics of each brain region are simulated with neural mass models and subsequently coupled together, considering the subject's empirical structural connectome. The parameters of the model can be optimized by comparing model-generated and empirical data. The resulting personalized whole-brain models have translational potential in neurosurgery, allowing investigators to simulate the effects of virtual therapies (such as resections or brain stimulations), assess the effect of brain pathology on network dynamics, or discern epileptic networks and predict seizure propagation in silico. The information gained from these simulations can be used as clinical decision support, guiding patient-specific treatment plans. Here the authors provide an overview of the rapidly advancing field of whole-brain modeling and review the literature on neurosurgical applications of this technology.

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KEYWORDS whole-brain model; computational neuroscience; functional neurosurgery; deep brain stimulation; epilepsy; neural mass model

A major goal of modern neurosurgery is the personalization of treatment approaches to optimize or predict outcomes for individuals.[1,2] Most work in this area has focused on genetic and molecular strategies, in which individual genetic or molecular profiles are used in diagnosis, treatment, and prediction of outcomes.[3–5] Advanced neuroimaging, such as diffusion-weighted imaging (DWI) and functional magnetic resonance imaging (fMRI), have also been used clinically to personalize treatment strategies. For example, DWI and fMRI are performed before neurooncological or epilepsy surgery to localize white matter tracts and eloquent cortex.[6–8] The surgeon uses this information to tailor resection strategies by considering the location of these regions. DWI has also been used to personalize targeting during deep brain stimulation (DBS).[9,10] The rationale for this is the idea that activation of specific white matter tracts may be involved in the clinical response to stimulation.[11] However, a major limitation of these presurgical mapping approaches is that they only provide a static snapshot of the brain, while ignoring the dynamic component that is important for the functioning of the brain in health and disease.[12,13] Furthermore, they provide no information with respect to how brain dynamics or functional architecture may change because of surgical transgression or stimulation of the identified tracts/regions. This may limit the ability to predict surgical outcomes, particularly for cognitive functions, where the relationship between specific white matter tracts or regions of the cortex and function is unclear. For example, measures of functional network topography have been shown to be important for cognitive function in patients with brain tumor,[14,15] and the preservation of functional network topography is increasingly considered to be a goal of neurooncological resection.[15–17] In DBS sur-

ABBREVIATIONS BOLD = blood oxygen level–dependent; DBS = deep brain stimulation; DWI = diffusion-weighted imaging; EEG = electroencephalography; ERP = event-related potential; fMRI = functional MRI; SEEG = stereotactic EEG; STN = subthalamic nucleus; VEP = Virtual Epileptic Patient.

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gery there is an implicit assumption that stimulation of the desired tract is critical due to the modulation of network activity. However, it is unclear how network dynamics will change in each individual patient. Therefore, it can be argued that DBS should be optimized with respect to the desired network dynamics rather than electrode location. This discrepancy has been observed in DBS for treatment-resistant depression.[18] Early work suggested that DBS leads positioned at the confluence of the forceps minor, cingulum bundle, uncinate fasciculus, and frontostriatal fibers would lead to a clinical response.[10] However, a prospective trial using DWI to target this region showed that these tracts were neither necessary nor sufficient for clinical response.[18] This discrepancy could be due to individual differences in brain network dynamics resulting from stimulation.

To understand the impact of surgical intervention on a patient's brain dynamics, computational simulations can be performed. The simulation is run by embedding a computational model in every network node of a patient's structural connectome. These models consist of a set of differential equations that determine the population-averaged activity of a single brain region according to the rationale of mean-field theory.[19] They are coupled according to the connectivity of the structural connectome and determine the evolution of dynamics across the whole brain.[20–22] These in silico models of brain activity allow for simulated perturbations of network elements and the analysis of downstream effects on brain dynamics. They hold translational promise, with the possibility of performing virtual resections, stimulations, and exploration and discovery of neurophysiological changes in diseased brains.

The development and availability of open-source neuroinformatics platforms such as The Virtual Brain[20,21] and the large-scale digital neuroscience research infrastructure EBRAINS[23] have made it feasible to produce personalized whole-brain models for clinical use. The objective of the current paper was to provide neurosurgeons with an overview of whole-brain modeling and to review the literature on neurosurgical applications of this technology.

Overview of Whole-Brain Modeling

Computational Modeling of Neural Populations

Mathematical modeling of nervous system function has an illustrious history, exemplified by the Nobel Prize–winning work of Hodgkin and Huxley.[24] Through detailed neurophysiological recordings of the squid giant axon, they derived a mathematical model of how action potentials are initiated and propagated. Although this is considered the canonical description of single-neuron activity, its utility in understanding whole-brain activity and resultant complex behavioral phenomena has been limited. Movement, perception, and cognition arise from the collective activity of interconnected neuronal populations, rather than from the activity of single neurons.[25–28] Furthermore, most in vivo empirical data on human brain functioning (e.g., electroencephalography [EEG], fMRI, and magnetoencephalography) reflect the collective activity of neuronal populations. To this end, neural population models have been developed to provide a mathematical framework representing the collective activity within an ensemble of neurons.[29,30] This approach is akin to the statistical physics approach taken when quantitatively describing the bulk properties of materials. Just as it is computationally inefficient and unnecessary to simulate the activity of every particle in a gas to track its temperature, it may be inefficient and unnecessary to simulate every neuron to understand the emergent properties of whole-brain activity.[31] These approximations of neural population activity are described as neural mass models and can be conceptualized as physiological or phenomenological models (although a continuum exists between these categories).[31,32] Physiological neural mass models can be considered a bottom-up approach to understanding neural population dynamics.

The differential equations that define these models incorporate state variables and parameters representing physiological properties of neurons (e.g., firing rates, membrane potentials, synaptic strengths and kinetics, and conduction delays). These neural mass models often exhibit topologies consisting of several conductance-based spiking excitatory neuron populations coupled to passive inhibitory subpopulations (Fig. 1). Common examples include the Wilson-Cowan[33] and the Jansen-Rit[34] models. Phenomenological models are considered top-down approaches, in which differential equations are defined to capture the empirically observed dynamics of large-scale neural activity, without as much grounding in biophysical concepts. The rationale for this approach is the observation that complex systems can exhibit specific behavior at different scales of organization that are not easily predicted from the activity of their constituent parts. One prominent example is the Kuramoto[35] model, which explains the neuronal population activity without differentiating neuronal subpopulation types. The state variable modeled is the phase of a limit cycle oscillator, and large-scale neuronal activity is described as the synchronization dynamics of multiple weakly coupled phase oscillators. Despite the abstract nature of these models, they reproduce complex synchronization behavior and other nonlinear dynamics observed from empirical neuroimaging data.

Whole-Brain Modeling

Neural mass models represent the dynamics emerging from a local population of interacting neurons at the mesoscale. However, these local populations do not function in isolation but are interconnected with many other local populations across the brain. Macroscale brain dynamics are therefore a function of both the local populations and the various interactions between them. Whole-brain models simulate these macroscopic brain dynamics by coupling many neural mass models across the brain according to an empirically defined connectivity architecture.[20,32,36] This takes advantage of advances in the analysis of diffusion MRI data, which allow for detailed, personalized descriptions of whole-brain connectivity (i.e., the connectome). A connectome can be described as a series of nodes and edges, where nodes represent cortical or subcortical regions and edges represent the connections between each region. Neural mass models are implemented at each node

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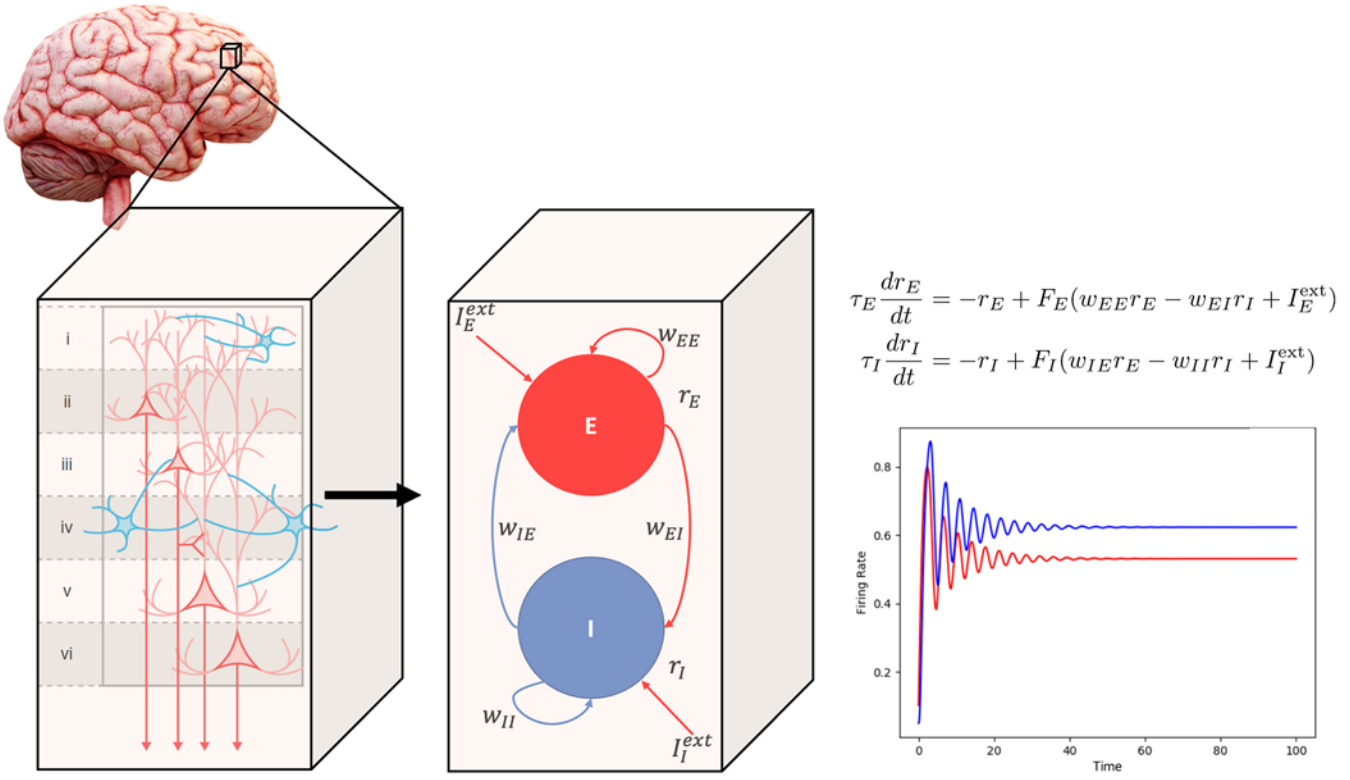


FIG. 1. An example of a neural mass model. In this example, a Wilson-Cowan neural mass model approximates a region of the cortex modeled as a cortical column consisting of interacting excitatory and inhibitory populations of neurons. The dynamics (average firing rate over time [in milliseconds]) of these populations are simulated through differential equations. These equations consider parameters such as connectivity between subpopulations, input and recurrent activity, and refractory periods. Figure is available in color online only.

of the connectome, and the degree of coupling between each is inferred from the weighting of the edges (e.g., streamline counts between each region). Time delays for signal propagation between nodes are determined by the Euclidean distance traversed by the streamlines, as well as a constant conduction speed (e.g., 5–18 msec/m).[37] A canonical whole-brain equation[20] considers these features and produces the spatiotemporal dynamics seen in whole-brain models:

$$x_i(t+1) = x_i(t) + f\big(x_i(t)\big)dt + c\sum_{j=1}^{N} w_{ij}\,\big(t - \Delta t_{ij}\big)dt + \eta(t),$$

where x_i is the mean field potential arising from the set of differential equations describing the neural mass model, t is the time course of the network mean field potential ($x_i(t)$) at node i (of N connected neural populations). Connection weights between nodes j and i (w_{ij}) are derived from fiber tract sizes/streamline count. Time delays for signal propagation ($\Delta t_{ij} = d_{ij}/v$) depend on a length matrix (d_{ij}) and a constant conduction speed (v). The term c is a scaling factor applied to the weight matrix. Finally, noise is a critical element in brain dynamics and is specified by $\eta(t)$. The specific type of noise depends on the specific model.[38]

By incorporating these properties of individual brains, the model operates within a biologically plausible space–time configuration and can generate realistic dynamic brain activity patterns. In reality, the neural states generated are hidden and are observed only through specific sensors (e.g., blood oxygen level–dependent [BOLD] signals or EEG potentials). For this reason, forward models must map the source space (simulated neural activity) into the sensor space to compare them with empirical data.[39,40] Parameters of the model are subsequently optimized by comparing model-generated with empirical data, yielding individually optimized models.[41] Parameter estimation has typically been evaluated by assessing all possible combinations of some constrained parameter space. Recently, novel deep learning–based approaches have been proposed to better optimize the model parameters while reducing the challenges due to large network sizes and the need for long simulation runs.[42] Figure 2 outlines the whole-brain simulation approach. Table 1 defines key terms.

Historically, the implementation of these models required knowledge of dynamical systems theory, advanced MRI analysis methodology, substantial programming ability, and access to high-performance computing resources. This limited their application and study to dedicated computational neuroscientists. The recent development of large-scale, open-source neuroinformatic platforms now allows for more accessibility to whole-brain modeling. For example, The Virtual Brain[20,21] can be used directly on EBRAINS[23] through a web graphical user in-

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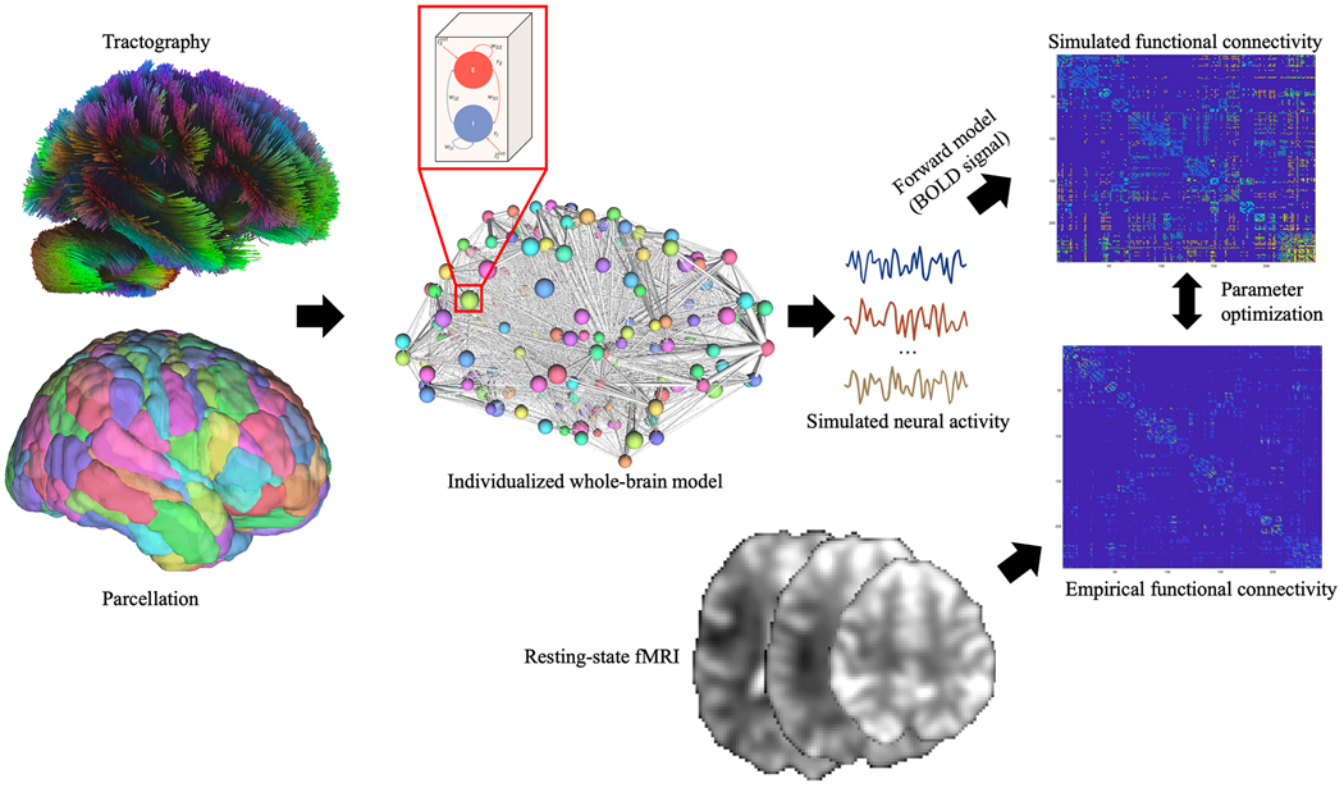


FIG. 2. Overview of an individualized whole-brain model pipeline. In vivo estimation of the structural connectome is accomplished with tractography and brain parcellation. Streamlines are counted and averaged according to the predefined parcellation, creating both a weight and length matrix to be used in model creation. Neural mass models are implemented at each node and a whole-brain equation is implemented to simulate the neural activity of the coupled system. Forward models are used to project the source activity to a specific sensor space. In this example, a hemodynamic model converts the neural population activity into a BOLD time series. Simulated functional connectivity can then be estimated and compared to empirically collected data. A parameter sweep can be performed to best fit the simulated data with the empirical data, resulting in individually optimized models. Optimized models can be used in neurosurgical simulations. Figure is available in color online only.

TABLE 1. Definitions of key terms

Term	Definition
Structural connectome	Description of the brain's structural connectivity as a graph. It is defined as a series of nodes & edges, where nodes represent cortical parcellations & edges represent a measure of structural connections, such as streamline counts using diffusion tensor imaging.
Functional connectome	Description of the brain's functional connectivity as a graph. It is defined as a series of nodes & edges, where nodes represent cortical parcellations & edges represent a measure of functional connections, such as BOLD time series correlation using fMRI.
Neural mass model	A system of differential equations describing the temporal evolution of neural population activity. These can be biophysically inspired using neuron motifs or they can describe phenomenological dynamics of population activity w/o grounding in specific biophysical concepts.
Whole-brain model	A group of neural mass models realized w/in the spatial scale of a human brain, giving rise to spatially & temporally realistic brain dynamics. Individual neural mass models are implemented at the nodes of the structural connectome & coupled together according to the weight & length of the edges.
Source model	The neural source activity generated by the whole-brain model.
Forward model	An approach that starts w/ modeling the source activity of a system (e.g., the neural activity), & then calculating the experimentally measurable signals that would arise from this behavior, like extracellular spikes, the local field potential, the EEG, or the BOLD signal.
Inverse model	A modeling approach that starts w/ an experimentally measured signal, & aims to infer aspects of the underlying behavior of the system that generated the measured signal.
Mean field theory	A mathematical approach suited for the statistical description of the collective behavior of a dynamical system comprising many interacting particles. In the case of neural circuits, mean field theory is typically used to provide a concise description of the firing rates of distinct but homogeneous subgroups of neurons.

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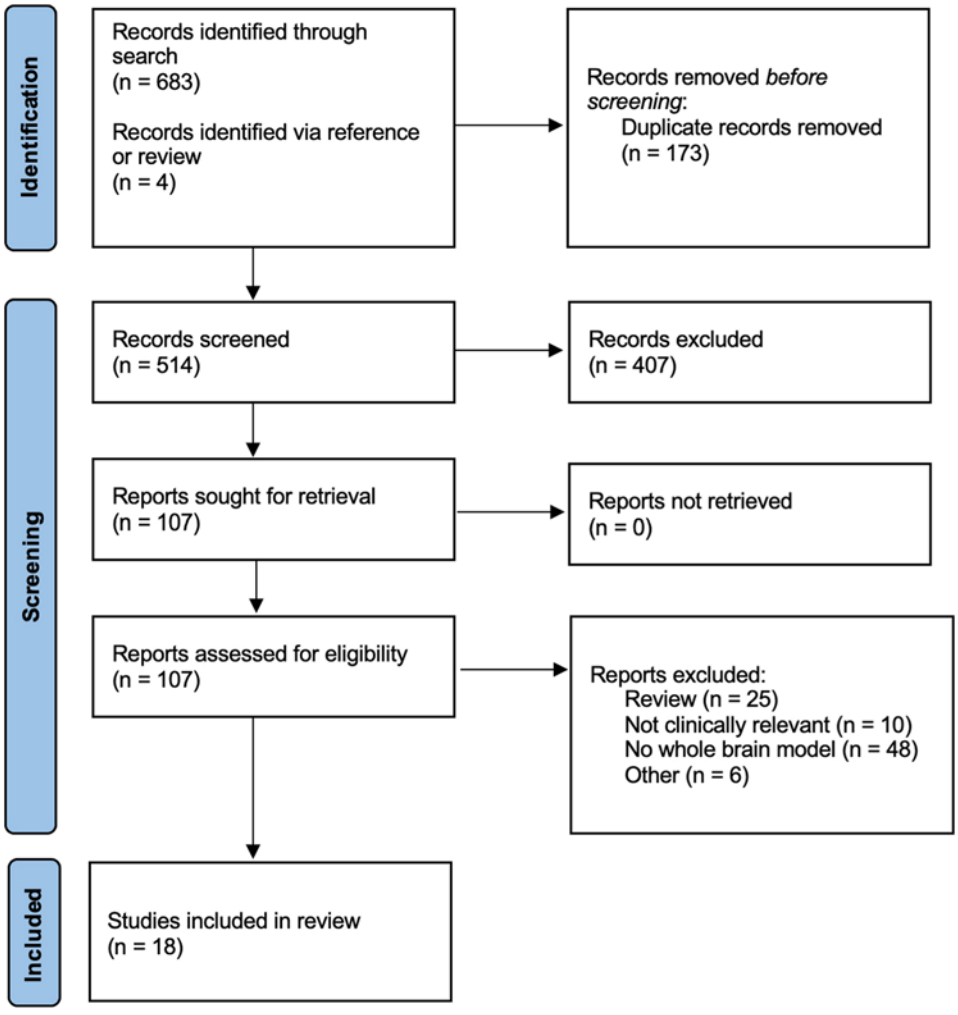


FIG. 3. PRISMA flowchart. Data added to the PRISMA template (from Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* 2021;372:n71) under the terms of the Creative Commons Attribution (CC BY 4.0) License (https://creativecommons.org/licenses/by/4.0/). Figure is available in color online only.

terface, without the need to install specific software and dependencies. The web interface connects with a network of supercomputers, allowing for rapid simulations that require extensive computational processing and storage space. Although producing and interpreting meaningful results still requires knowledge about image analysis and modeling, these services have lowered the barrier of entry into this burgeoning field, directly impacting the ability to investigate whole-brain models for clinical purposes.

Methods

In this section, we present a review of the literature focusing on neurosurgical applications of whole-brain models. We queried PubMed, Scopus, and Web of Science with the search terms ((“whole brain modelling”) OR (“computational modelling”) OR (“the virtual brain”)) OR (“Epileptor”) OR (“personalized model”) OR (“virtual epileptic patient”) AND ((“neurosurgery”) OR (“epilepsy”) OR (“brain tumor”) OR (“deep brain stimulation”)). Our query was performed on April 20, 2023, and included papers published up until this time. Studies were reviewed by the lead author. They were included if they created the whole-brain model as coupled neural mass models implemented according to the connectome architecture of the human brain and focused on a clinical topic in the fields of neurooncology, epilepsy, or DBS. Studies were excluded if the computational model was built on a subsystem of the brain (rather than the whole brain), the whole-brain model was constructed differently (e.g., without neural mass models or without considering the connectome), if the clinical context was outside the defined areas of interest, or if they were primarily technical reports. The reference lists of relevant articles were inspected to identify studies missed by the search strategy. The PRISMA flowchart is displayed in Fig. 3 and a descriptive summary of the included studies can be found in Table 2. Each included article was examined for the number of subjects, type of neural mass model, and type of simulated data. Due to the diversity of studies, the results were summarized in a descriptive manner.

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TABLE 2. Literature review of articles on modeling of whole-brain dynamics

Authors & Year	Subjects	Model	Simulated Data	Result
Neuro-oncology				
Aerts et ., 2018[44]	25 preop pts w/ brain tumor	Reduced Wong-Wang model w/ individually optimized parameters	BOLD time series & functional connectivity	Prediction of empirical functional connectivity improved w/ individually optimized parameters. Local model parameters can differentiate tumor-infiltrated regions from healthy regions. Associations btwn model parameters & cognition.
Aerts et al., 2020[45]	25 pre- & postop pts w/ brain tumor	Reduced Wong-Wang model w/ individually optimized parameters	BOLD time series & functional connectivity	Optimized model parameters are stable following tumor resection. Proof-of-concept demonstrating postop empirical brain dynamics can be predicted w/ virtual neurosurgery.
Epilepsy				
Taylor et al., 2015[49]	Single pt w/ DRE	Amari-type neural field	Source (neural activity)	Optimal stimulation control algorithm can prevent sz onset.
Hutchings et al., 2015[50]	22 pts w/ temporal lobe epilepsy	Bi-stable phenomenological model	Source (neural activity)	Pt-specific resections, as compared to standard clinical resections, may improve clinical outcomes.
Jirsa et al., 2017[55]	Single pt w/ bitemporal epilepsy & hypothalamic hamartoma	Epileptor model	SEEG	First description of VEP as a generative model of sz dynamics using individualized structural connectome. Good correspondence of spatiotemporal sz evolution btwn empirical & simulated data. VEP captures simple & complex sz spread. Allows for discovery of spatial distribution of epileptogenicity parameters.
Proix et al., 2017[59]	15 pts w/ DRE	Epileptor model	SEEG	Individualized virtual brain models reliably predict sz propagation zone. Worse clinical outcome was observed in cases in which predicted propagation zones were not included in clinical hypothesis & resection.
An et al., 2019[56]	7 pts w/ DRE	Epileptor model	Source (neural activity)	Graph analysis of structural connectomes & whole-brain simulations identified surgical target zones outside the epileptic zone that minimize sz propagation & maintain normal brain dynamics.
Olmi et al., 2019[65]	15 pts w/ DRE	Epileptor model	SEEG	Minimal & selective lesioning of unstable connections in the epileptic zone, identified via linear stability analysis, can halt sz propagation in the model.
Courtiol et al., 2020[79]	14 pts w/ DRE & 5 healthy controls	Extended Epileptor model	BOLD time series	Epileptic brains demonstrate lower global excitability in the interictal state w/ locally increased excitability w/in epileptic zones.
Vattikonda et al., 2021[64]	25 pts w/ DRE	Epileptor model	SEEG	Precision of model predicted epileptic zone location is high in sz-free pts (0.75) & low in non–sz-free pts (0.4).
Makhalova et al., 2022[58]	53 pts w/ DRE	Epileptor model	SEEG	VEP-derived epileptic zone showed moderate concordance w/ the epileptogenic zone identified w/ epileptogenicity index. VEP predictions had better concordance w/ clinical outcomes. In non–sz-free pts, there was a higher no. of regions defined as epileptogenic by the VEP that remained nonresected.
Dollomaja et al., 2023[80]	Single pt w/ status epilepticus	Epileptor	SEEG	VEP reproduced sz dynamics in status epilepticus.
Wang et al., 2023[60]	53 pts w/ medically refractory epilepsy	Epileptor	SEEG	VEP reproduced the clinically defined epileptic zone w/ a precision of 0.6. VEP showed lower false discovery rates in sz-free pts compared to non–sz-free pts.
DBS				
van Hartevelt et al., 2014[68]	Single pt w/ PD	Dynamic mean field model	BOLD time series & functional connectivity	Modeling before & 5 mos after DBS revealed a shift in brain dynamics toward a healthy dynamical regime. Suggested recovery of structural connectivity w/ long-term DBS.

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TABLE 2. Literature review of articles on modeling of whole-brain dynamics

Authors & Year	Subjects	Model	Simulated Data	Result
van Hartevelt et al., 2015[69]	Single pt w/ PD	Dynamic mean field model	BOLD time series & functional connectivity	Changing the connectivity weights of the STN w/ the thalamus & putamen resulted in optimal fitting of simulated postop functional connectivity w/ empirical postop structural connectivity. This suggests STN DBS results in Hebbian plasticity of connections btwn the STN & thalamus/putamen.
Saenger et al., 2017[70]	10 pts w/ PD	Supercritical Hopf bifurcation model	BOLD time series & functional connectivity	Therapeutic DBS shifted brain dynamics toward a healthy dynamical state. Exploration of in silico DBS ranked nodes by the amount they pushed the dynamical regime toward a healthy state. Top nodes included the thalamus, putamen, & hippocampus.
An et al., 2022[71]	3 pts w/ Tx-resistant depression	Generic 2D oscillator	EEG & ERPs	Simulation of selective fiber tract activations from individualized electrode contact positioning resulted in distinct stimulus-induced ERPs. Parameter optimization resulted in good correspondence of simulated & empirical data for some, but not all, stimulation locations. Provides a model for investigating the individual network effects of DBS.
Meier et al., 2022[74]	Normative structural connectivity	Spiking model for basal ganglia & reduced Wong-Wang mean field model for cortex	Spike rates	Description of a multiscale model, incorporating whole-brain dynamics w/ a spiking model of the basal ganglia. DBS of GPi & STN increased firing rates in thalamus, insula, & frontal regions, w/ a slight decrease in the SMA. Multiscale model increases the biological realism of simulation & allows for investigation of network effects of DBS.

DRE = drug-resistant epilepsy; GPi = globus pallidus internus; PD = Parkinson's disease; pts = patients; SMA = supplementary motor area; Sz = seizure; Tx = treatment.

Results

Neurooncology

The interaction between gliomas and functional and structural brain networks is well recognized.[43] Correspondingly, resection of these lesions is increasingly considered within the context of connectomics,[16,43] with the goal of preserving network structure and function. Virtual brain models allow for the simulation of brain activity in the context of brain tumors and for performing virtual resections. For example, the structural connectome can be altered systematically to simulate a resection and the resulting brain dynamics can be simulated on the altered connectome (Fig. 4). Ultimately, the goal is to use the simulated brain dynamics to predict outcome measures for individuals.

In a proof-of-concept study, Aerts et al.[44] demonstrated the feasibility of simulating large-scale brain dynamics in the context of brain tumors. The authors used reduced Wong-Wang models[36] (dynamic mean field neural mass model consisting of one excitatory and one inhibitory neural population) coupled according to each subject's empirical structural connectome. A hemodynamic forward model was used to simulate BOLD time series data, and the simulated data were compared to empirical fMRI data. Parameters of the model were optimized to maximize the correlation between simulated and empirical functional connectivity, and these individually optimized models were then used for further analysis. The authors found that individualized tuning of model parameters improved the prediction accuracy of empirical functional connectivity, reaching an average linkwise correlation of 0.33 across subjects. Furthermore, they demonstrated that local model parameters can differentiate regions directly affected by a tumor versus regions distant from the tumor. Specifically, they found that median local inhibitory connection strengths were lower and more variable in tumor regions compared to nontumor regions and healthy control regions. Non–tumor-affected regions in patients had higher local inhibitory strengths compared to healthy control brains. These results suggest that the presence of a tumor has distance-dependent effects on local inhibitory connection weights. This is consistent with local hyperexcitability found near glial tumors and could theoretically be used to delineate resection targets.

In a second paper,[45] the same authors used these individually optimized virtual brain models to perform virtual resection and simulate postoperative brain dynamics. Simulated postoperative functional connectivity was calculated and compared with the empirical functional connectivity obtained from postoperative scanning. First, they found that model parameters remained similar before and after actual tumor resection. This suggests that optimized preoperative models can be used to simulate postoperative dynamics. Next, they performed virtual neurosurgery on the preoperative models. For this, a tumor resection mask was used to identify cortical regions and white matter tracts that were transgressed by the surgery. The identified nodes and edges were then removed from the preoperative model and the resultant brain dynamics were simulated on this altered connectome. They found that virtual neurosurgery and the resulting simulated functional connectivity resulted in improved prediction of empirical functional

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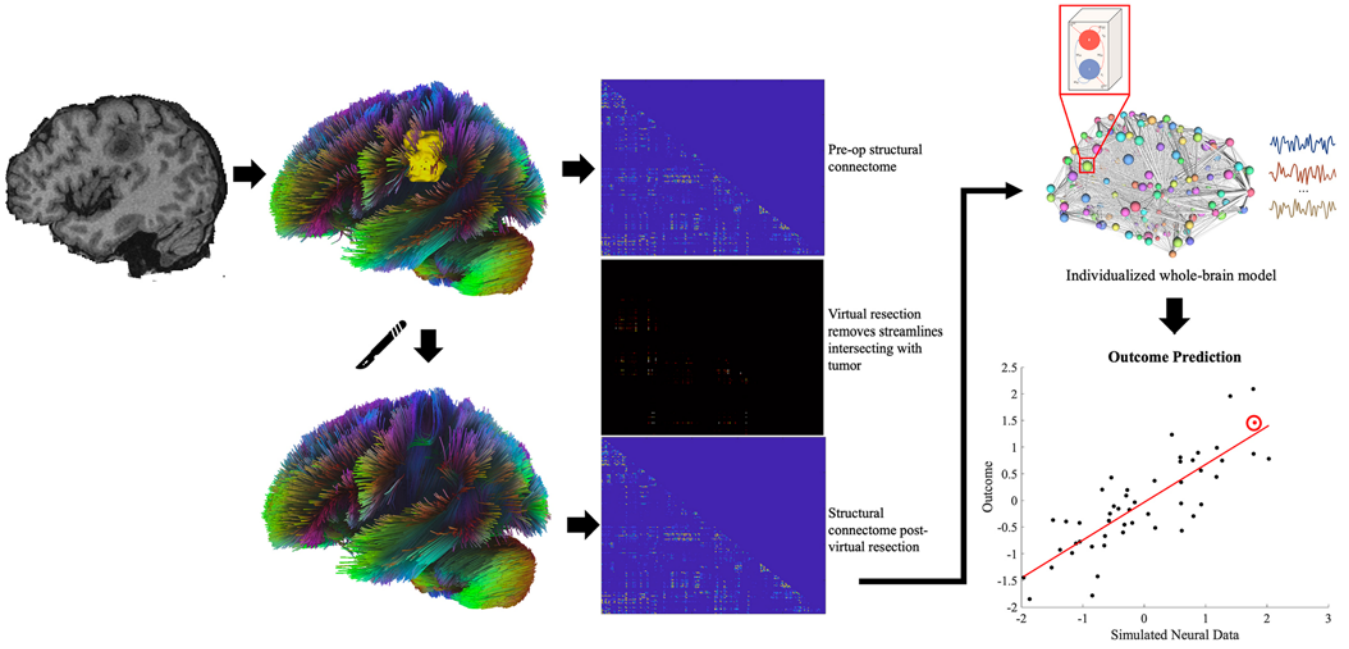


FIG. 4. An example of a virtual resection. The structural connectome is computed in the presence of the patient's brain tumor (*yellow region*). Streamlines intersecting the tumor are virtually resected, leaving an altered connectome. Whole-brain modeling is subsequently implemented using the altered connectome, resulting in simulated brain dynamics representing the postoperative state. The simulated brain dynamics can be analyzed and used for the prediction of individual outcome measures. Figure is available in color online only.

connectivity in a subset of patients. The lesioned structural connectome alone better predicted postoperative functional connectivity in other patients, rather than simulated functional connectivity. The authors suggested that in these patients, brain dynamics are determined principally by the underlying structure. This work demonstrated the feasibility of performing virtual neurosurgery in patients with brain tumors. It remains to be seen if these models will have clinical utility in neurooncology, and this should be the focus moving forward.

Epilepsy

As a dynamic network disease of the nervous system, epilepsy is ideally suited for computational modeling. Indeed, there are many mathematical models, ranging from microscopic models of ion channel dynamics in animal brain slices to whole-brain mean field models of seizure propagation in human brains.[46,47] In contrast to other applications, epilepsy simulations typically use disease-specific models rather than models of general neural activity. These disease models typically consist of a bistable system (in which a stochastic process shifts dynamics into a stable seizure or nonseizure attractor state) or a system with parameter-induced bifurcations (in which a slowly changing parameter moves the system from a fixed-point attractor representing a seizure-free state, onto a limit cycle representing a seizure state).[48] Within the neurosurgical context, the goals of these models include identifying epileptic zone and propagation zone regions, performing virtual interventions, and predicting patient outcomes. In an early example, Taylor et al.[49] used a whole-brain model of bistable epilepsy dynamics to develop an optimal control algorithm that was used to drive the system from a seizure state into a nonseizure state. This work suggests that neurostimulation paradigms can be designed to abate seizures by inputting a stimulus at a critical point to prevent the trajectory of the system from entering a pathological state. Hutchings et al.[50] performed virtual resections in patient-specific models of temporal lobe epilepsy. Consistent with the clinical literature,[51,52] virtual amygdalohippocampectomy reduced the likelihood of seizures in 72.7% of patients. However, by performing patient-specific virtual resections based on an individualized computational model of seizure origin, 100% of patients had reduced seizure likelihood. This study is limited in that the nodes commonly removed during the patient-specific virtual resections are not feasible targets for removal (e.g., thalamus), although they may be amenable to neurostimulation strategies. Overall, the work suggests that patient-specific models can be used to delineate individualized surgical targets to improve clinical outcomes.

Most of the studies using whole-brain models in epilepsy research have used a phenomenological neural mass model named the Epileptor.[53,54] It characterizes seizure dynamics by using a subsystem responsible for fast discharges, a second subsystem generating spike and wave events, and a slowly evolving state variable that links these two subsystems and controls the transition to and from ictal states. When Epileptors were coupled according to a patient's structural connectome, epileptic seizure dynamics recorded by intracranial EEG could be accurately reproduced.[55] This model was termed the Virtual Epileptic Patient (VEP) and has been used to investigate individualized epileptic zones/propagation zones and to predict

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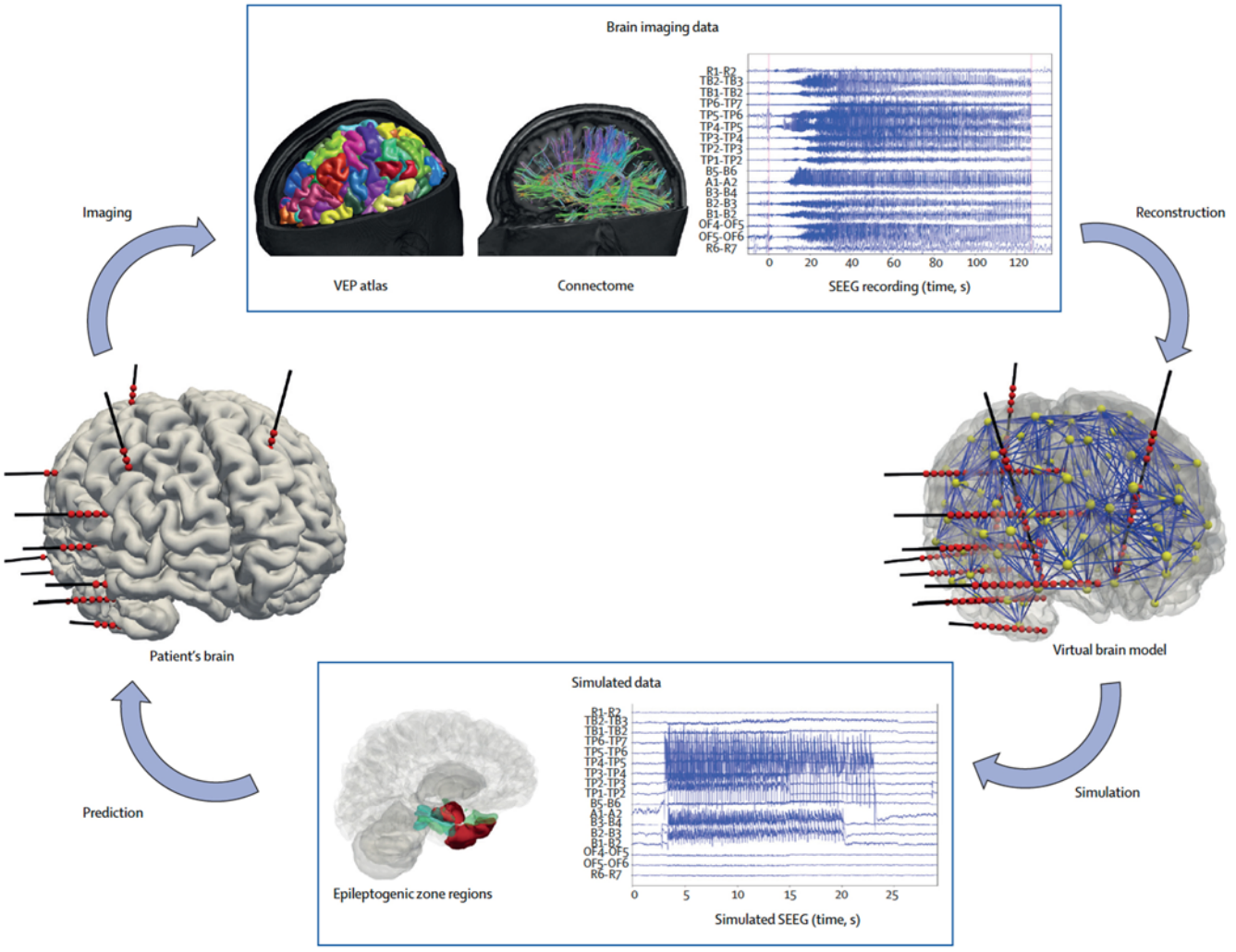


FIG. 5. Modeling and inference workflow of the VEP. T1-weighted MRI was done to obtain brain geometry data and to define distinct brain regions according to the VEP atlas (*upper left*). Tractography was done to estimate the length and density of white matter fibers from diffusion-weighted MRI (*blue lines* in the virtual brain model), which establish the connectome that specifies the connection strength and time delays via signal propagation between two brain regions. After implantation of SEEG electrodes (*red spheres* and *black lines* depicted over the patient's brain and the virtual brain model), intracranial recordings were obtained and coregistered in the same reference space and with the exact locations of the SEEG electrodes. This set of structural brain imaging data (the VEP atlas and connectome) was combined to reconstruct a virtual brain model. The virtual brain model supplies the computational neuronal source activity model for each brain region (*yellow spheres*) in the VEP atlas, connects the brain regions through the connectome (*blue lines*), and computes the measurable SEEG sensor signal for the implantation (*red spheres* and *black lines*). Machine learning and artificial intelligence methods are used to estimate the model parameters, including epileptic zone approximation, and to complete the patient-specification process of the virtual brain model. The virtual brain model can be used to simulate the consequences of different epileptic zone hypotheses (*red* and *green areas* show epileptic zone regions) and corresponding SEEG recordings, and these simulated data can subsequently be used to predict brain activity and behavior under various conditions. Reprinted from *The Lancet Neurology*, Vol. 22, Jirsa et al., Personalised virtual brain models in epilepsy, pp 443-454, Copyright 2023,[61] with permission from Elsevier. https://www.sciencedirect.com/journal/the-lancet-neurology. Figure is available in color online only.

surgical outcomes[55–61] (Fig. 5). To reduce uncertainty and optimize key parameters, the model considers prior information about the spatial distribution of the epileptic zone from clinical investigations. Bayesian inference methods are used for model inversion, resulting in epileptogenicity values for each brain region.[62–64]

Proix et al.[59] validated the VEP by demonstrating that it could predict the propagation zone as determined empirically by stereotactic EEG (SEEG) and clinical expertise. The model identified propagation zone regions that were unexplored by SEEG, and therefore not considered in the clinical hypothesis. A higher number of unexplored propagation zone regions was related to worse seizure outcomes, suggesting that the reason for surgical failure was an inadequate clinical hypothesis of the seizure network. This work was extended by comparing VEP predictions to a state-of-the-art clinical analysis incorporating the epileptogenicity index.[58] The authors found that epileptic zones

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predicted by the VEP had strong concordance with clinical analysis. However, patients with poor outcomes following surgery had a higher number of regions defined as the epileptic zone exclusively by the VEP, again suggesting that these regions were missed by the clinical analysis.

Wang et al. reproduced these findings in a cohort of 58 patients undergoing resection.[60] The VEP had lower false discovery rates in seizure-free patients (mean 0.028) compared with non–seizure-free patients (mean 0.407). False discovery epileptic zone predictions in seizure-free patients suggest a high possibility of the predicted area actually being outside the epileptic network, whereas false positives in non–seizure-free patients suggests a high possibility of nonresected epileptic regions. The results suggest that the predictive power of the VEP could be better exploited in the non–seizure-free patients. Building on this work, the model is now being investigated in a multicenter prospective trial of epilepsy surgery outcomes (ClinicalTrials.gov identifier: NCT03643016).

The VEP has also been used to investigate alternative surgical strategies for medically refractory epilepsy. Olmi et al.[65] used personalized whole-brain models of seizure dynamics to demonstrate that seizure propagation can be halted via selective lesioning of key connections originating in the epileptic zone. This suggests that minimally invasive, targeted disconnections may be as effective as removal of the entire epileptic zone. An et al.[56] used the VEP to identify alternative surgical targets for cases in which the epileptic zone is located in eloquent cortex and therefore cannot be resected. The identified target zones minimize seizure propagation in each patient, suggesting that virtual brain simulations may help guide surgical treatment strategies in cases otherwise deemed inoperable.

Deep Brain Stimulation

Increasing evidence suggests that the clinical effects of DBS result from the modulation of brain networks and alteration of brain dynamics.[9,66,67] Whole-brain modeling studies give insight into DBS mechanisms at the large-scale network level and can simulate network dynamics resulting from stimulation. Taking advantage of a scenario in which a patient with DBS received both pre- and postoperative DWI, van Hartevelt et al.[68] used a whole-brain computational model to investigate the network effects of chronic stimulation. They found that chronic stimulation may alter the structural connectome, resulting in a shift in neural dynamics toward a pattern observed in healthy controls. Using the same model, they demonstrate that the subthalamic nucleus (STN) may undergo Hebbian plasticity changes with the putamen and the thalamus.[69] Saenger et al.[70] also used whole-brain models to investigate brain dynamics in patients with Parkinson disease undergoing DBS. Using a supercritical Hopf bifurcation model, they found that therapeutic DBS shifted brain dynamics toward a critical oscillatory regime similar to that seen in healthy brains. Furthermore, they performed in silico DBS to rank potential stimulation targets by the amount they could drive brain dynamics toward this healthy regime. Identified nodes included the thalamus, globus pallidus, orbitofrontal cortex, and hippocampus, suggesting that these may be ideal targets for maximal modulation of whole-brain dynamics.

In a study using The Virtual Brain platform, An et al.[71] validated computational brain models against empirical stimulation event-related potentials (ERPs) recorded in three patients receiving DBS for treatment-resistant depression. The cortical surface was reconstructed as a triangular mesh and individual 2D oscillators were placed at each vertex, coupled according to the individual structural connectome. Virtual stimulation was applied to fiber tracts intersecting each electrode contact, and a forward model was used to project the resulting source activity into sensor (EEG) space. The simulation results demonstrated distinct ERPs as a function of exact stimulation location. These showed good correspondence with empirical data for some stimulation positions, but not for others. A major limitation of this study was that the DWI scans used to create the models were obtained 6 months before surgery, whereas the empirical ERP data were obtained 3 years after surgery. Any DBS-induced plasticity changes would therefore have been neglected. Despite this, the study demonstrates the feasibility of in silico DBS with highly detailed and personalized whole-brain models, opening up the possibility of investigating and optimizing the network effects of DBS in individuals. In future work, PET, fMRI, and/or magnetoencephalography acquired during stimulation could be used to validate the predictions arising from these models.

Limitations

Assumptions and simplifications are built into whole-brain models at each stage of formation. This includes the construction of the structural connectome (reliant on DWI processing and tractography algorithm choice); choice of brain parcellation (assumptions about the organization of the brain); the implementation of neural mass models (assumptions about the dynamic nature of neural populations); and parameter selection (can be arbitrary). Highlighting this is the fact that even highly detailed and optimized models only moderately predict empirical data, achieving a correlation in the range of 15%–50%.[45] The inclusion of cortical surface reconstructions,[57,71,72] multiscale models that integrate multiple levels of spatial resolution,[73,74] and neurotransmitter/receptor density maps that modulate local and global dynamics[75] will be important in creating more realistic models. Moreover, these models make a key assumption about the human brain: that macroscale network dynamics play a mechanistic role in human behavior and disease pathology, and that by modeling the former (network dynamics) we can learn something about the latter (human behavior/disease pathology). Although this assumption is supported by a large body of work, it remains to be shown whether modeling these phenomena will translate into clinically useful tools. Many of the identified papers describe models built on a small number of patients. Larger studies using more patient data will be required to ensure the generalizability of the results. Given that the goal of this article was to provide a broad overview of the field, we primarily used a narrative approach to summarizing the literature. We believe that our search strategy has identified all relevant articles

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emphasizing neurosurgical applications of personalized whole-brain models that use empirical connectivity and neural mass models. However, studies examining whole-brain models that are constructed in an alternative fashion will have been excluded. Furthermore, we focused on areas with clear surgical relevance. Other related clinical areas, such as traumatic brain injury,[76] stroke,[77] and neurodegenerative disease[78] have also been studied using whole-brain models, but are not within the scope of this article.

Conclusions

Parallel advancements in neuroimaging and large-scale open-source neuroinformatic platforms are expediting the clinical translation of personalized whole-brain models. These have translational potential in neurosurgery, allowing for the investigation of virtual resections, virtual stimulations, and the exploration of neurophysiological changes in diseased brains. Besides the opportunities to use these models as research platforms, the simulation of brain dynamics may be used as clinical decision support tools by neurosurgeons. In this article we have provided an accessible overview of these models that will hopefully inspire interest in the community so that their utility can be fully realized.

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Author Contributions

Conception and design: Lang. Acquisition of data: Lang. Analysis and interpretation of data: Lang. Drafting the article: Lang, Griffiths. Critically revising the article: all authors. Reviewed submitted version of manuscript: Lang, Momi, Vetkas, Santyr, Yang, Kalia, Lozano. Approved the final version of the manuscript on behalf of all authors: Lang. Study supervision: Griffiths, Lozano.

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